Online Examination Report

Date:	28.07.2014
Attempts:	1
Examinee:	Atwoki Kaija, Albert
Date of birth:	
Subject:	020-AGK
Result:	0 %
Time used:	00:00:22 (HH:MM:SS)

Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

Attachment 1: List of result by question

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
1	2769	21.	AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	0,00	1,00
2	2827	21.	AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	0,00	1,00
3	2751	21.	AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	0,00	1,00
4	1577	21.2.3	Aeroplane: Wings, tail and control surfaces	0,00	1,00
5	1122	21.4	LANDING GEAR, WHEELS, TYRES, BRAKES	0,00	1,00
6	2348	21.6.3	Aeroplane: Pressurisation and air conditioning system	0,00	1,00
7	2926	21.8	FUEL SYSTEM	0,00	1,00
8	2250	21.9.1	General, definitions, basic applications: etc	0,00	1,00
9	814	21.9.1.6	Electromagnetism	0,00	1,00
10	817	21.9.3.1	DC Generation	0,00	1,00
11	239	21.9.3.1	DC Generation	0,00	1,00
12	26	21.9.3.2	AC Generation	0,00	1,00
13	379	21.10.1.2	Engine: design, operation etc	0,00	1,00
14	2440	21.10.10.1	Performance	0,00	1,00
15	2241	21.11.1	Basic principles	0,00	1,00
16	1437	21.11.1.1	Basic generation of thrust, the thrust formula	0,00	1,00
17	308	21.11.1.2	Design, types of turbine engines, components	0,00	1,00
18	1851	21.11.4	Engine Operation and Monitoring	0,00	1,00
19	1001	21.12.2.2	Fire detection	0,00	1,00
20	1506	21.13	OXYGEN SYSTEMS	0,00	1,00
21	2801	22.	INSTRUMENTATION	0,00	1,00
22	4122	22.	INSTRUMENTATION	0,00	1,00
23	2636	22.	INSTRUMENTATION	0,00	1,00
24	119	22.2.1	Pressure measurement	0,00	1,00
25	972	22.2.1	Pressure measurement	0,00	1,00
26	990	22.2.1	Pressure measurement	0,00	1,00
27	792	22.2.1.1	Definitions	0,00	1,00
28	1265	22.2.4	Altimeter	0,00	1,00

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
29	1140	22.2.6	Airspeed indicator (ASI)	0,00	1,00
30	107	22.2.6	Airspeed indicator (ASI)	0,00	1,00
31	1294	22.2.7	Mach meter	0,00	1,00
32	1646	22.3	MAGNETISM - DIRECT READING COMPASS AND FLUX VALVE	0,00	1,00
33	1587	22.3.3	Direct Reading Magnetic Compass	0,00	1,00
34	1595	22.4	GYROSCOPIC INSTRUMENTS	0,00	1,00
35	1461	22.4.2	Rate of turn and slip indicator/Turn Co-ordinator	0,00	1,00
36	226	22.4.5	Remote reading compass systems	0,00	1,00
37	1422	22.6.3	Flight Director : design and operation.	0,00	1,00
38	852	22.8.1	Trim systems : design and operation.	0,00	1,00
39	2165	22.12.3	Stall Warning Systems (SWS)	0,00	1,00
40	1802	22.12.8	Radio Altimeter	0,00	1,00
			Total:0%	**0,00**	**40,00**

Attachment 2: List of result by topic

Licence
Date: 28.07.2014
Examinee: Atwoki Kaija, Albert
Location: Uganda Civil Aviation Authority

Your Result: 0,00 from 40,00 Points (0,00 %)

Topics	Ammount of Questions	Points achieved	Maximum Points	Percent
21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	20	0,00	20,00	0,00
21.2. AIRFRAME	1	0,00	1,00	0,00
21.4. LANDING GEAR, WHEELS, TYRES, BRAKES	1	0,00	1,00	0,00
21.6. PNEUMATICS - PRESSU AND AIRCO SYSTEMS	1	0,00	1,00	0,00
21.8. FUEL SYSTEM	1	0,00	1,00	0,00
21.9. ELECTRICS	5	0,00	5,00	0,00
21.10. PISTON ENGINES	2	0,00	2,00	0,00
21.11. TURBINE ENGINES	4	0,00	4,00	0,00
21.12. PROTECTION AND DETECTION SYSTEMS	1	0,00	1,00	0,00
21.13. OXYGEN SYSTEMS	1	0,00	1,00	0,00
22. INSTRUMENTATION	20	0,00	20,00	0,00
22.2. MEASUREMENT OF AIR DATA PARAMETERS	8	0,00	8,00	0,00
22.3. MAGNETISM - DIRECT READING COMPASS AND FLUX VALVE	2	0,00	2,00	0,00
22.4. GYROSCOPIC INSTRUMENTS	3	0,00	3,00	0,00
22.6. AEROPLANE : AUTOMATIC FLIGHT CONTROL SYSTEMS	1	0,00	1,00	0,00
22.8. TRIM, Y/D, FLIGHT ENVELOPE PROTECTION	1	0,00	1,00	0,00
22.12. ALERTING SYSTEMS, PROXIMITY SYSTEMS	2	0,00	2,00	0,00
Total	**40**	**0,00**	**40,00**	**0,00**

1

Running a fuel tank dry before switching tanks is not a good practice because

- ☐ any foreign matter in the tank will be pumped into the fuel system.
- ☐ the engine-driven fuel pump or electric fuel boost pump draw air into the fuel system and cause vapor lock.
- ☐ the engine-driven fuel pump is lubricated by fuel and operating on a dry tank may cause pump failure.

ID:	2769(LTS:222960, LTM:1086)	Max. Points	1,00
Syllabus reference:	21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	Points	0,00

2

What energy source is used to drive the turbine of a turbocharged airplane?

- ☐ **Ignition system.**
- ☐ **Engine exhaust gases.**
- ☐ **Engine compressor.**

ID:	2827(LTS:222975, LTM:1101)
Syllabus reference:	21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT
Max. Points	1,00
Points	0,00

3

One advantage of fuel injection systems over carburetor systems is

- ☐ **less difficulty with hot weather vapor locks during ground operations.**
- ☐ **better fuel distribution to the cylinders.**
- ☐ **easier hot-engine starting.**

4

In a steep turn to the left, when using spoilers ...

- ☐ **The right aileron will descend, the left one will ascend, the right spoiler will retract and the left one will extend.**
- ☐ **The right aileron will descend, the left one will ascend, the right spoiler will extend and the left one will retract.**
- ☐ **The right aileron will ascend, the left one will descend, the right spoiler will retract and the left one will extend.**
- ☐ **The right aileron will ascend, the left one will descend, the right spoiler will extend and the left one will retract.**

ID:	1577(LTS:222614, LTM:371)	Max. Points	1,00
Syllabus reference:	21.2.3 Aeroplane: Wings, tail and control surfaces	Points	0,00

5

A main landing gear is said to be "locked down" when:

- ☐ the corresponding indicator lamp is amber.
- ☐ the strut is locked by an overcentre mechanism.
- ☐ the actuating cylinder is at the end of it's travel.
- ☐ it is in the down position.

ID:	1122(LTS:222494, LTM:250)	Max. Points	1,00
Syllabus reference:	21.4 LANDING GEAR, WHEELS, TYRES, BRAKES	Points	0,00

6

The cabin pressure is regulated by the:

- ☐ **Cabin inlet airflow valve.**
- ☐ **Air cycle machine.**
- ☐ **Air conditioning pack.**
- ☐ **Outflow valve.**

ID:	2348(LTS:222832, LTM:958)	Max. Points	1,00
Syllabus reference:	21.6.3 Aeroplane: Pressurisation and air conditioning system	Points	0,00

7

While in flight, ice begins forming on the outside of the fuel tank in use. This could indicate

- ☐ a broken dip tube.
- ☐ water in the fuel.
- ☐ a leak in the fuel line.

ID:	2926(LTS:222986, LTM:1123)	Max. Points	1,00
Syllabus reference:	21.8 FUEL SYSTEM	Points	0,00

8

A Constant Speed Drive aims at ensuring

- ☐ that the starter-motor maintains a constant RPM not
- ☐ that the electric generator produces a constant frequency.
- ☐ equal AC voltage from all generators.
- ☐ that the CSD remains at a constant RPM not withstanding the generator RPM withstanding the acceleration of the engine.

ID: 2250(LTS:222791, LTM:928)
Syllabus reference: 21.9.1 General, definitions, basic applications: etc

Max. Points 1,00
Points 0,00

9

If a current is passed through a conductor which is positioned perpendicular to a magnetic field:

- ☐ *the current will increase.*
- ☐ *the intensity of the magnetic field will decrease.*
- ☐ *there will be no effect unless the conductor is moved.*
- ☐ *a force will be exerted on the conductor.*

10

The reason for using inverters in an electrical system is

- ☐ To change DC into AC.
- ☐ To change AC into DC.
- ☐ To avoid a short circuit.
- ☐ To change the DC voltage.

ID: 817(LTS:222420, LTM:176)
Syllabus reference: 21.9.3.1 DC Generation

Max. Points 1,00
Points 0,00

11

The type of windings commonly used in DC starter motors are:

- ☐ *shunt wound.*
- ☐ *compound wound.*
- ☐ *series wound.*
- ☐ *series shunt wound.*

ID:	239(LTS:222295, LTM:51)	Max. Points	1,00
Syllabus reference:	21.9.3.1 DC Generation	Points	0,00

12

The purpose of a voltage regulator is to control the output voltage of the:

- ☐ ***batteries at varying loads.***
- ☐ ***TRU.***
- ☐ ***generator at varying loads and speeds.***
- ☐ ***generators at varying speeds and the batteries at varying loads.***

ID:	26(LTS:222250, LTM:7)	Max. Points	1,00
Syllabus reference:	21.9.3.2 AC Generation	Points	0,00

13

In general, in twin-engine aeroplanes with 'constant speed propeller'

- ☐ the oil pressure turns the propeller blades towards smaller pitch angle.
- ☐ the spring force turns the propeller blades towards smaller pitch angle.
- ☐ the oil pressure turns the propeller blades towards higher pitch angle.
- ☐ the aerodynamic force turns the propeller blades towards higher pitch angle.

ID:	379(LTS:222322, LTM:78)	Max. Points	1,00
Syllabus reference:	21.10.1.2 Engine: design, operation etc	Points	0,00

14

Under what condition is V(MC) the highest?

- ☐ **CG is at the most rearward allowable position.**
- ☐ **CG is at the most forward allowable position.**
- ☐ **Gross weight is at the maximum allowable value.**

ID: 2440(LTS:222882, LTM:1019)
Syllabus reference: 21.10.10.1 Performance

Max. Points 1,00
Points 0,00

15

In a turbo-jet, the purpose of the turbine is to ...

- ☐ **clear the burnt gases, the expansion of which provide the thrust**
- ☐ **drive devices like pumps, regulator, generator.**
- ☐ **compress the air in order to provide a better charge of the combustion chamber**
- ☐ **drive the compressor by using part of the energy from the exhaust gases**

ID:	2241(LTS:222788, LTM:925)	Max. Points	1,00
Syllabus reference:	21.11.1 Basic principles	Points	0,00

16

Regarding a jet engine:
I. The maximum thrust remains constant as the pressure altitude increases.
II. The specific fuel consumption increases slightly as the pressure altitude increases at constant TAS.

- ☐ ***I is incorrect, II is correct.***
- ☐ ***I is correct, II is correct.***
- ☐ ***I is incorrect, II is incorrect.***
- ☐ ***I is correct, II is incorrect.***

ID:	1437(LTS:222575, LTM:332)	Max. Points	1,00
Syllabus reference:	21.11.1.1 Basic generation of thrust, the thrust formula	Points	0,00

17

The bypass ratio:
I. is the ratio of bypass air mass flow to HP compressor mass flow.
II. can be determined from the inlet air mass flow and the HP compressor mass flow.

- ☐ ***I is incorrect, II is incorrect.***
- ☐ ***I is correct, II is correct.***
- ☐ ***I is incorrect, II is correct.***
- ☐ ***I is correct, II is incorrect.***

ID:	308(LTS:222309, LTM:65)	Max. Points	1,00
Syllabus reference:	21.11.1.2 Design, types of turbine engines, components	Points	0,00

18

Low oil pressure is sometimes the result of a

- ☐ restricted oil passage
- ☐ worn oil pump
- ☐ too small scavenger pump.
- ☐ too large oil pump

ID:	1851(LTS:222708, LTM:456)	Max. Points	1,00
Syllabus reference:	21.11.4 Engine Operation and Monitoring	Points	0,00

19

Smoke detectors fitted on transport aircraft are of the following type :

- ☐ optical or ionization
- ☐ magnetic
- ☐ electrical
- ☐ chemical

ID: 1001(LTS:222471, LTM:218)
Syllabus reference: 21.12.2.2 Fire detection

Max. Points 1,00
Points 0,00

20

The disadvantages of a chemical oxygen source for the passenger cabin are : 1. a flow which cannot be modulated, 2. a heavy and bulky system, 3. non reversible functioning, 4. risks of explosion, 5. poor autonomy. The combination regrouping all the correct statements is:

- ☐ **1, 2, 3, 4, 5**
- ☐ **1, 2, 3, 5**
- ☐ **1, 3, 5**
- ☐ **2, 4**

ID: 1506(LTS:222595, LTM:352)
Syllabus reference: 21.13 OXYGEN SYSTEMS

Max. Points 1,00
Points 0,00

21

What should be the indication on the magnetic compass as you roll into a standard rate turn to the right from a south heading in the Northern Hemisphere?

- ☐ **The compass will initially indicate a turn to the left.**
- ☐ **The compass will remain on south for a short time, then gradually catch up to the magnetic heading of the airplane.**
- ☐ **The compass will indicate a turn to the right, but at a faster rate than is actually occurring.**

ID:	2801(LTS:222968, LTM:1094)	Max. Points	1,00
Syllabus reference:	22. INSTRUMENTATION	Points	0,00

See Attachments:

1. instrument_84.jpg Figure 84 (Attachment No.1)

22

(Refer to Figure 84.) Which altimeter depicts 8,000 feet?

- ☐ 2
- ☐ 3
- ☐ 1

ID:	4122(LTS:223032, LTM:1169)	Max. Points	1,00
Syllabus reference:	22. INSTRUMENTATION	Points	0,00

23

A gyroplane will have the greatest tendency to roll during

- ☐ climbing flight in which forward airspeed decreases.
- ☐ descending flight in which forward airspeed decreases.
- ☐ horizontal flight at high speed.

ID: 2636(LTS:222937, LTM:1074)
Syllabus reference: 22. INSTRUMENTATION

Max. Points 1,00
Points 0,00

24

An aircraft is flying from a cold air mass into a warm air mass. The TAS and true altitude will:

- ☐ **Both increases.**
- ☐ **TAS decreases, true altitude increases.**
- ☐ **TAS increases, true altitude decreases.**

ID:	119(LTS:222269, LTM:25)	Max. Points	1,00
Syllabus reference:	22.2.1 Pressure measurement	Points	0,00

25

If the static pressure ports iced over while descending from altitude, the airspeed indicator would read:

- ☐ **High**
- ☐ **Correctly.**
- ☐ **Low**

ID:	972(LTS:222465, LTM:212)	Max. Points	1,00
Syllabus reference:	22.2.1 Pressure measurement	Points	0,00

26

An airfield, elevation 3000 feet, has a pressure altitude of 3500 feet. What is the QFE?

- ☐ **913.s hPa**
- ☐ **879.8 hPa**
- ☐ **896.5 hPa**

ID:	990(LTS:222467, LTM:214)	Max. Points	1,00
Syllabus reference:	22.2.1 Pressure measurement	Points	0,00

27

Given:
Pt = total pressure
Ps = static pressure
Pso = static pressure at sea level

The CAS is a function of:

- [ ] ***Pt - Pso***
- [ ] ***(Pt - Pso) / Ps***
- [ ] ***Pt - Ps***
- [ ] ***Pt / Ps***

ID:	792(LTS:222415, LTM:171)	Max. Points	1,00
Syllabus reference:	22.2.1.1 Definitions	Points	0,00

28

If the static source of an altimeter becomes blocked during a descent the instrument will:

☐ *under-read.*

☐ *gradually indicate zero.*

☐ *continue to display the reading at which the blockage occurred.*

☐ *indicate a height with the subscale setting on QNH.*

ID: 1265(LTS:222531, LTM:287)
Syllabus reference: 22.2.4 Altimeter

Max. Points 1,00
Points 0,00

29

An ASI consists of a gas tight casing which has:

- ☐ static pressure fed outside a sealed capsule
- ☐ static pressure fed through a restrictive choke
- ☐ pitot pressure fed into an open capsule and static pressure fed outside the capsule
- ☐ static pressure fed outside a sealed capsule and pitot pressure fed into a second open capsule

ID:	1140(LTS:222498, LTM:255)	Max. Points	1,00
Syllabus reference:	22.2.6 Airspeed indicator (ASI)	Points	0,00

30

An aircraft is maintaining FL 120 in cloud. The ASI reading falls to zero. The most probable cause is:

- ☐ **Static vent blocked by ice.**
- ☐ **ASI malfunction.**
- ☐ **Pitot head and static vent blocked by ice.**

ID:	107(LTS:222268, LTM:24)	Max. Points	1,00
Syllabus reference:	22.2.6 Airspeed indicator (ASI)	Points	0,00

31

If an aircraft climbs at a constant IAS in the standard atmosphere, the Mach number will:

- ☐ **increase**
- ☐ **remain constant**
- ☐ **remain constant unless the aircraft flies through an inversion**
- ☐ **decrease**

ID:	1294(LTS:222537, LTM:294)	Max. Points	1,00
Syllabus reference:	22.2.7 Mach meter	Points	0,00

32

The rotor of the DGI spins up and away from the pilot when 090 is indicated. The latitude compensation nut situated on the near right hand side of the inner gimbal from the gyro axis, has been set to give zero drift on the ground at the equator. To compensate for earth rotation at 30 S the latitude compensating nut:

- ☐ **must be adjusted inwards;**

- ☐ **is not adjusted since the latitude nut can only be used to correct for apparent wander in the northern hemisphere.**

- ☐ **must be adjusted outwards;**

ID:	1646(LTS:222642, LTM:389)	Max. Points	1,00
Syllabus reference:	22.3 MAGNETISM - DIRECT READING COMPASS AND FLUX VALVE	Points	0,00

33

The turning error of a direct reading magnetic compass:

- ☐ *increases when the magnetic latitude increases.*
- ☐ *does not depend on the magnetic latitude.*
- ☐ *decreases when the magnetic longitude increases.*
- ☐ *decreases when the magnetic latitude increases.*

ID:	1587(LTS:222617, LTM:374)	Max. Points	1,00
Syllabus reference:	22.3.3 Direct Reading Magnetic Compass	Points	0,00

34

The principle of rigidity is used for the operation of the following gyroscopic instruments:

- ☐ **Directional Gyro and Artificial Horizon.**
- ☐ **Artificial Horizon and Turn indicator.**
- ☐ **Directional Gyro and Turn indicator.**

ID:	1595(LTS:222618, LTM:375)	Max. Points	1,00
Syllabus reference:	22.4 GYROSCOPIC INSTRUMENTS	Points	0,00

35

What is the approximate angle of bank for a rate one turn at 110 knots?

- ☐ **25 degrees**
- ☐ **30 degrees**
- ☐ **18 degrees**

ID:	1461(LTS:222582, LTM:339)	Max. Points	1,00
Syllabus reference:	22.4.2 Rate of turn and slip indicator/Turn Co-ordinator	Points	0,00

36

A gyromagnetic compass consists of:

1 - a horizontal axis gyro
2 - a vertical axis gyro
3 - an earth's magnetic field detector
4 - an erection mechanism to maintain the gyro axis horizontal
5 - a torque motor to make the gyro precess in azimuth

The combination that regroups all of the correct statements is:

- ☐ ***1, 4.***
- ☐ ***2, 3.***
- ☐ ***2, 3, 5.***
- ☐ ***1, 3, 4, 5.***

ID: 226(LTS:222293, LTM:49)
Syllabus reference: 22.4.5 Remote reading compass systems

Max. Points 1,00
Points 0,00

37

The horizontal command bar of a flight director:

1 - repeats the position information given by the ILS in the horizontal plane
2 - repeats the position information given by the ILS in the vertical plane
3 - gives information about the direction and the amplitude of the corrections to be applied on the pitch of the aircraft.

The combination that regroups all of the correct statements is:

- ☐ ***2, 3.***
- ☐ ***2.***
- ☐ ***3.***
- ☐ ***1, 3.***

ID:	1422(LTS:222571, LTM:328)	Max. Points	1,00
Syllabus reference:	22.6.3 Flight Director : design and operation.	Points	0,00

38

The purpose of a trim tab (device) is to:

- ☐ trim the aeroplane at low airspeed.
- ☐ reduce or to cancel control forces.
- ☐ trim the aeroplane during normal flight.
- ☐ lower manoeuvring control forces.

ID: 852(LTS:222427, LTM:183)
Syllabus reference: 22.8.1 Trim systems : design and operation.

Max. Points 1,00
Points 0,00

39

Which of these signals are inputs, at least, in the stall warning computers?

- ☐ **Angle of attack and flaps and slats deflection.**
- ☐ **Angle of attack, flaps deflection, EPR and N1.**
- ☐ **Angle of attack, flaps deflection and EPR.**
- ☐ **Angle of attack and flaps and spoilers deflection.**

ID:	2165(LTS:222766, LTM:903)	Max. Points	1,00
Syllabus reference:	22.12.3 Stall Warning Systems (SWS)	Points	0,00

40

The radio altimeter uses the following wavelengths:

- ☐ *myriametric.*
- ☐ *metric.*
- ☐ *millimetric*
- ☐ *centimetric.*

ID: 1802(LTS:222683, LTM:440)
Syllabus reference: 22.12.8 Radio Altimeter

Max. Points 1,00
Points 0,00

Attachment No. 1

Questions 22

Figure 84

instrument_84.jpg

1

2

3

4

Figure 84. Altimeter/8,000 Feet © ASA